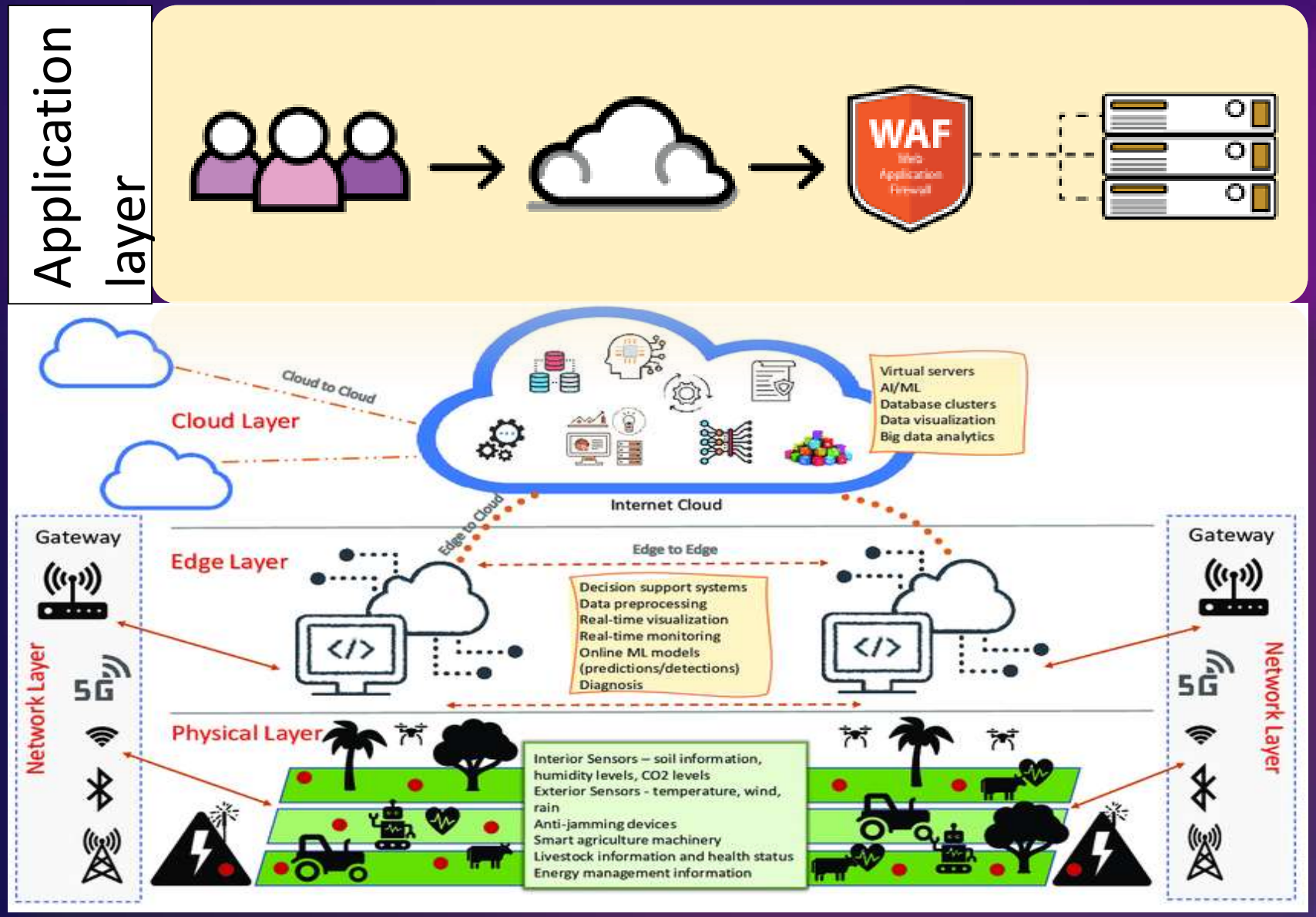
WIRELESS COMMUNICATION

BY SNEHA JADHAV

❖ SMART IRRIGATION:



❖ Sensors implemented in smart agriculture:

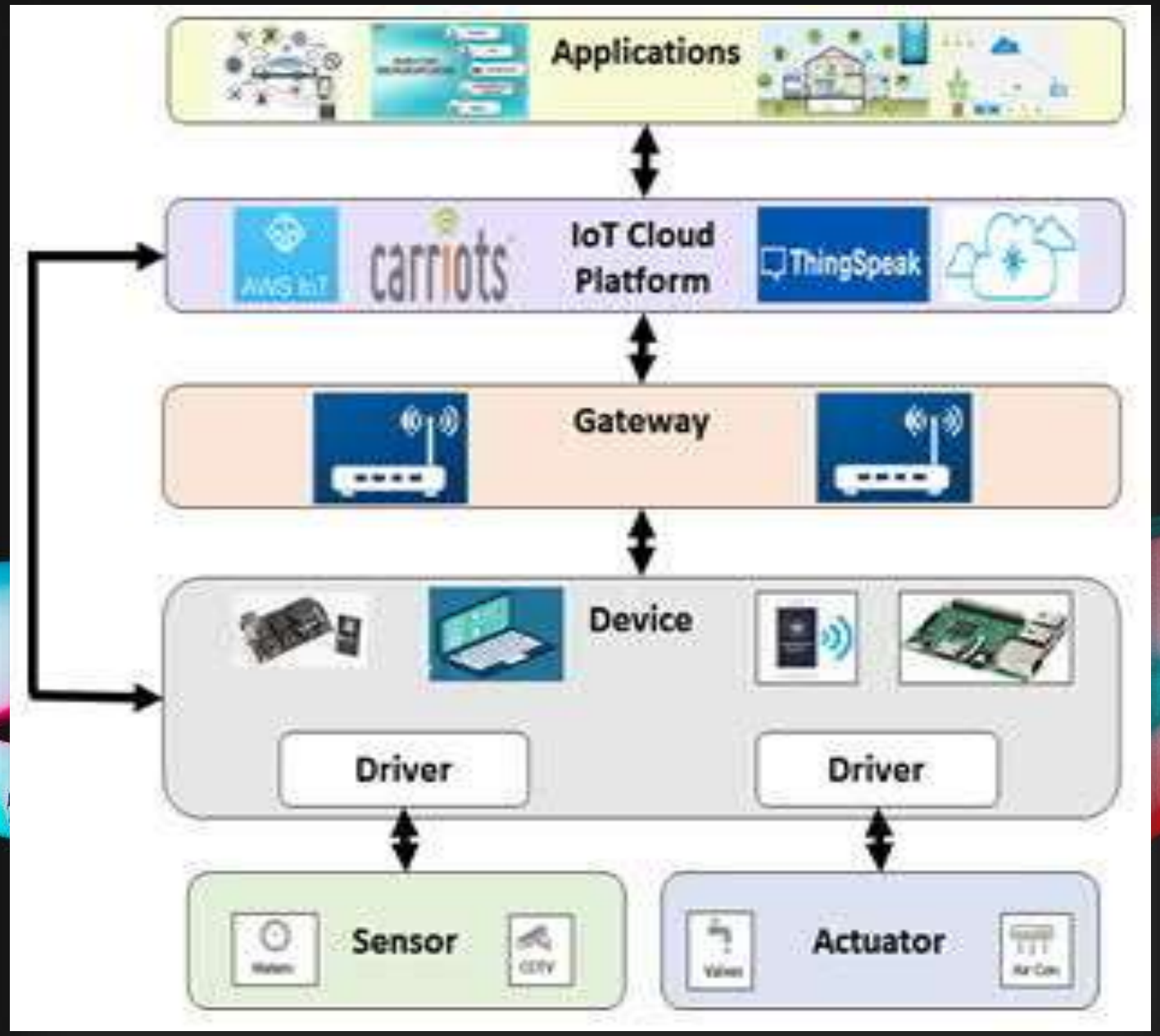
Optical Sensors In Agriculture

Electrochemical Sensors For Soil Nutrient Detection

Dielectric Soil Moisture Sensors

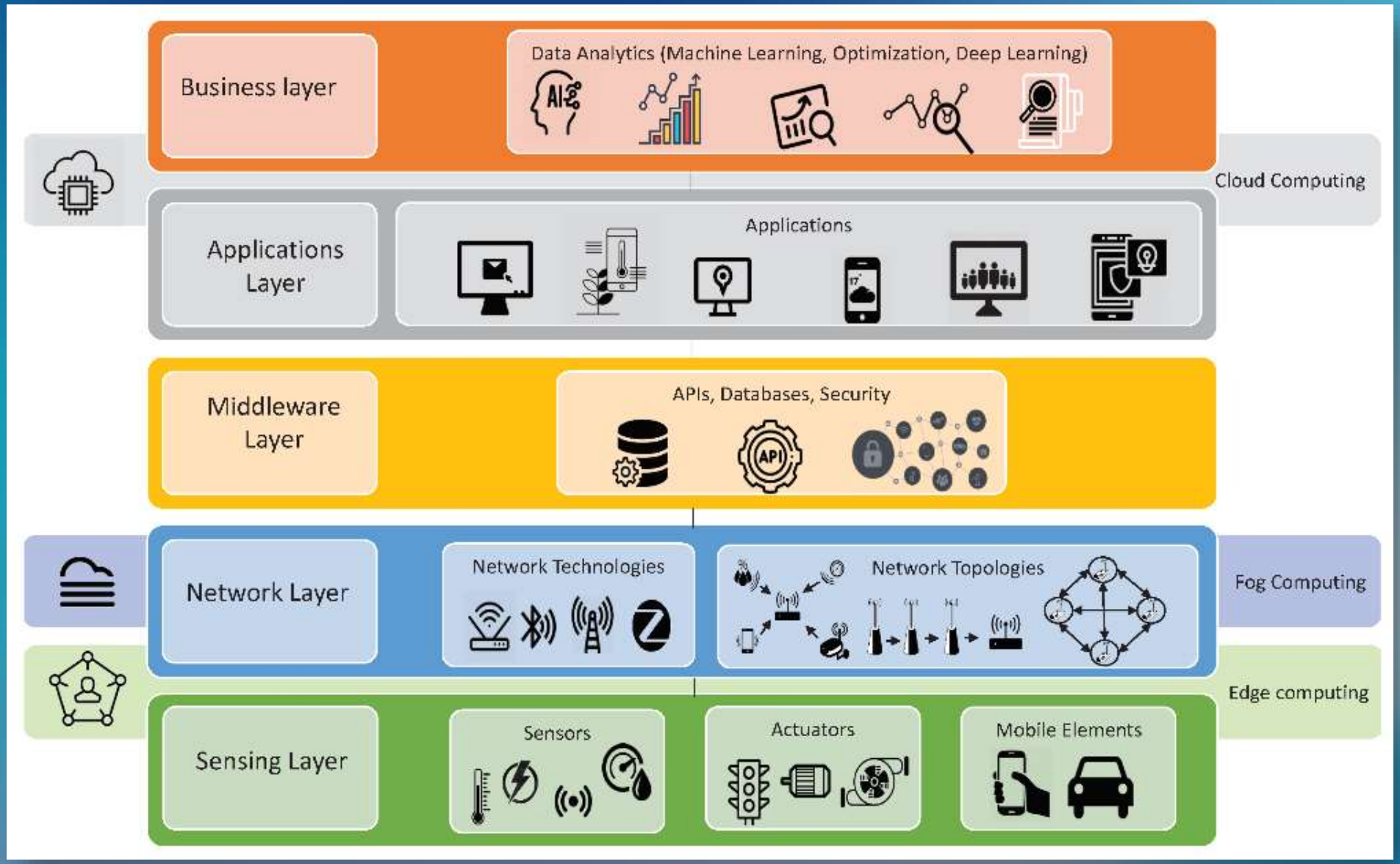
Location Sensors In Agriculture

❖ SMART HOME AUTOMATION:

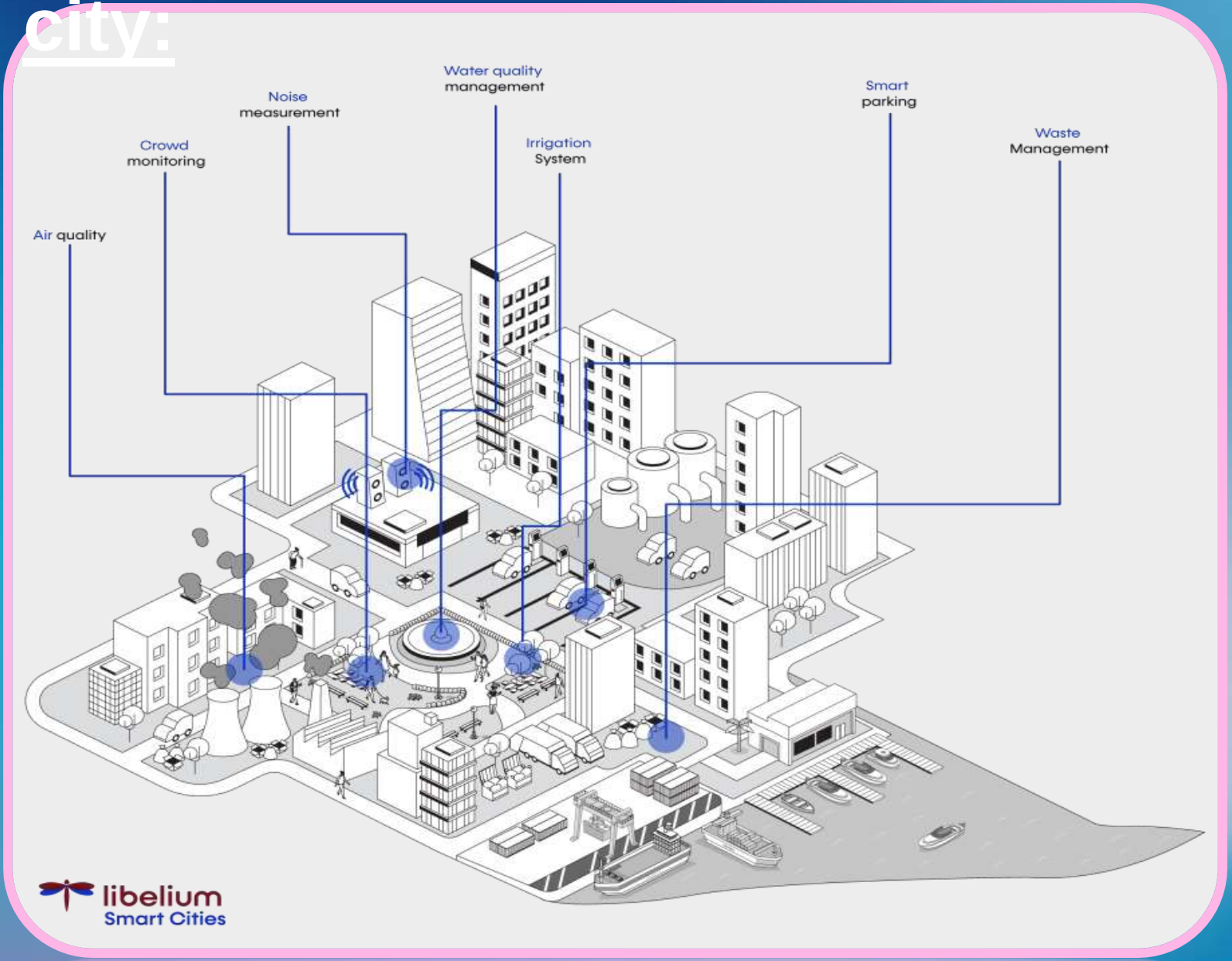


❖ Sensors implemented in smart home:

❖ SMART CITY:


Business layer
Data Analytics (Machine Learning, Optimization, Deep Learning)
Cloud Computing
Applications Layer
Applications
Middleware Layer
APIs, Databases, Security
Network Layer
Network Technologies
Network Topologies
Fog Computing
Edge computing
Sensing Layer
Sensors
Actuators
Mobile Elements

❖ Sensors implemented in smart city:



❖ What is LPWAN?

- LPWANs (Low-Power Wide-Area Networks) are a type of radio-based connectivity used for wireless data communication for IoT devices and M2M (machine-to-machine) applications.
- Features:

1. Confined or remote coverage
2. Power efficiency
3. Security and privacy

1G

- Year 1980.
- It has analog voice.
- Unable to transmit data

2G

- Year 1991.
- It has digital voice.
- Can send texts, SMS.
- Can use internet.

3G

- Year 1990.
- Emphasis on high speed wireless connection.
- Used for calling, video calling, and in playing games

4G

- Year 2002.
- Faster than 3G.
- Upgrade video quality, games full HD.
- Streaming more smooth.

5G

- Year 2014.
- Can connect to internet.
- 10 times faster than 4G.

❖ Comparision between 1G to 5G:

PARAMETERS	1G	2G	3G	4G	5G
TECHNOLOGY	Analog cellular	Digital cellular	CDMA/ IP protocol	IP and combo of LAN,WAN,WLAN,PAN	4G + www
BANDWIDTH	2kbps	14-64kbps	2mbps	200mbps	>1gbps
CORE NETWORK	PSTN	PSTN	Packet network	Internet	Internet
MULTIPLEXING	FDMA	TDMA/CDMA	CDMA	CDMA	CDMA
SERVICES	Mobile telephony	Digital voice, short messaging	High quality audio, video, data	Dynamic information access, variable devices	Dynamic information access, variable devices

Bluetooth :

- A wireless technology for sending and receiving digital voice and data over the cellphone and other devices.
- Example: wireless mouse, Bluetooth headsets, etc

Wifi:

- Technology used to connect computers, tablets, smartphones and other devices on internet.
- A radio signal sent from wireless router to a nearby device which translates signal into data .

Zigbee:

- Is based on the institute of electrical and electronics engineering(IEEE).
- Works with digital radios by allowing different devices to converse through one another.

RFID:

- Refers to a wireless system which uses radio frequency of radio waves.
- It consists of 2 components RFID reader and an RFID tag.

❖ COMPARISION OF BLUETOOTH, WIFI,RFID,ZIGBEE

PARAMETERS	BLUETOOTH	WIFI	ZIGBEE	RFID
Standard	802.15.1	802.11	802.15.4	
Frequency band	2.4 GHz	2.4 GHz and 5GHz	2.4 GHz, 850 – 930 MHz	30 KHz to 500 KHz,
Data rate	1-3 Mbps	10-100+ Mbps	20-250 Kbps	100 Mbps
Transmission range	Up to 100m	Up to 100m	Up to 100m	3 to 6 meters
Power consumption	Very low	High	Low	Low
Network topology	Ad hoc, point to point, star	Point to hub, ad hoc	Mesh, star, tree, ad hoc	-
Application	Wireless audio streaming and data transfer, smart wearables and fitness trackers, beacon networks	Wireless local area network connection, broadband Internet access	Home automation and control, industrial monitoring sensor network	It is used in several commercial and industrial applications, keeping track of items checked out of a library.

❖ Comparision between LoRa, Sigfox, Nb-IOT, CAT-M

PARAMETERS	LORA	SIGFOX	NB-IOT	CAT-M
TECHNOLOGY	Chirp Spread Spectrum modulation	proprietary Ultra Narrow Band (UNB)	Cellular(LTE)	3GPP (LTE)
BANDWIDTH	250KHz	100Hz	200kHz	1.4 MHz to 5 MHz
RANGE	1km	10km	1km	10 km
POWER CONSUMPTION	low	low	Low	Low
APPLICATIONS	Industrial IOT, smart meter for gas, water, electricity	Smart parking, supply chain management, environment monitoring.	Smart metering,smart building,healthcar, logistic and supply chain	Smart agriculture, healthcare, security systems

Thank you